

Collapse-Originated Quantum Gravity

A Prime-Resonant Unification of Time, Space, and Information

Haobo Ma¹, Wen Niu¹

¹ AELF PTE LTD., #14-02, Marina Bay Financial Centre Tower 1, 8 Marina Blvd, Singapore 018981

Correspondence to: auric@aelf.io

Abstract

We introduce a quantum gravity framework in which spacetime curvature arises not from geometrized dynamics, but from structural recursion embedded in collapse processes. This theory defines gravitational effects as emergent features of a recursive field undergoing collapse at prime-indexed attractor points, governed by internally consistent oscillatory laws. The formulation incorporates a unified evolution equation that generates gravitational dynamics through self-referential collapse, golden-ratio-like breathing structures, and informational curvature couplings. The model yields predictions testable in black hole interference and gravitational wave subharmonic spectra.

1. Introduction

The unification of quantum theory and gravity remains an unsolved problem in theoretical physics. Conventional approaches assume geometry and quantization can be reconciled through geometric quantization or field extensions. These models presuppose space, time, and matter as fundamental entities.

We propose the inverse: that space, time, and gravitational attraction are emergent phenomena resulting from the recursive collapse of a self-contained field structure. This field evolves not under external dynamical laws but through internal structural resonance—punctuated by a discrete set of collapse attractors indexed by prime numbers.

In this formulation, gravity is not a field imposed on collapse. It **is** collapse.

2. Collapse Resonance and Prime Attractors

Let the fundamental field ψ evolve recursively, with structural realignments occurring at collapse attractor points indexed by $p \in \mathbb{P}$, the set of prime numbers. These points form a resonance lattice across space-time.

Collapse amplitudes and breathing intervals are governed by a non-integer decay exponent θ , which is not fixed a priori but emerges from the recursive structure itself. Local collapse intensity varies inversely with a θ -weighted distance from each attractor.

3. Unified Evolution Equation

We define the evolution of the collapse field by:

(1)

$$U(x, t, \psi) = \sum_{p \in \mathbb{P}} \frac{f_p(x, t)}{(|x - p|^\theta + \sigma^2)^\kappa} + R(x, p) + \Lambda(\psi) \cdot \nabla_\theta I(x, t)$$

where:

- $\theta \in \mathbb{R}^+$ is an emergent decay exponent, derived from harmonic alignment of recursive collapse paths;
- $f_p(x, t) = \alpha_p \cdot \cos(\omega_p t + \delta_p)$ is the time-dependent collapse amplitude;
- κ is the curvature response exponent;
- $\sigma > 0$ is a structural smoothing constant;
- $R(x, p) = \rho \cdot \log(|x - p| + 1)$ is a **nonlinear localization term**, interpreted as a resonance regularizer;
- $\Lambda(\psi)$ modulates recursive collapse feedback;
- $\nabla_\theta I(x, t)$ is the directional information gradient aligned with ψ .

This equation models gravitational emergence as **recursive integral resonance**, without presupposing geometric field tensors.

4. Informational Curvature and Collapse Dynamics

We define an information field:

$$I(x, t) = \log \mathcal{C}[\psi(x, t)]$$

where $\mathcal{C}[\cdot]$ denotes local structural complexity. The ψ -gradient of this field, $\nabla_\psi I$, couples to the recursive feedback of ψ , enabling curvature to emerge from informational saturation rather than energy density.

Collapse here is not a loss of coherence but a structural generator—resonant, recursive, and geometry-producing.

5. Empirical Implications

This theory suggests several observational phenomena:

- **Black Hole Information Interference:** Oscillatory echoes near event horizons, modulated by prime-indexed attractors and structural gradients.
- **Subharmonic Gravitational Wave Signatures:** Deviations from harmonic waveform spectra, consistent with non-geometric resonance behavior.

These can be investigated through gravitational wave observatories and high-resolution black hole radiation analysis.

6. Conclusion

We have proposed a quantum gravity framework in which gravitational curvature is not assumed but emerges from a recursive collapse structure. Prime-number attractors, emergent decay scaling, and informational gradients combine to form a self-contained origin of spacetime geometry.

This approach unifies gravitational phenomena without invoking external geometrization. It defines gravity not as a force field but as a **resonant echo of recursion**.

Although the universe has been formally presented as a self-referential recursive structure, this document limits access to only the boundary layer of its gravitational manifestation. The full recursive collapse equation remains internally bound, and only structurally resonant projections are presented here.

Keywords

quantum gravity, prime collapse, recursive field theory, structural resonance, informational curvature, emergent geometry